

BOOBALAN A

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Professional Summary

Data Scientist with end-to-end experience across the full data science lifecycle—from raw telematics data ingestion and EDA to feature engineering, machine learning model development, and production deployment on GCP. Experienced in geospatial intelligence, predictive analytics, and automated MLOps pipeline development within the commercial vehicle domain. Comfortable owning problems end-to-end, from SQL queries and data analysis to stakeholder-facing dashboards and productionized solutions.

Experience

VE Commercial Vehicles (VECV) | Data Scientist I **Mar 2025 – Present**
Geospatial & Telematics Analytics / GCP / Python / MLOps

- **Reducing False Positives in DSMS (Ongoing) — YOLO:** Developing a YOLO-based computer-vision pipeline to classify and filter false-positive alerts generated by the Driver Safety Monitoring System (DSMS) using image-based analysis; applying object detection on captured frames to validate alert triggers, improving alert precision and reducing false alarm rates for fleet managers.
- **Geospatial Clustering Engine (HAC + Haversine):** Designed and deployed a production geospatial model that identifies high-frequency vehicle stoppage zones at 500m precision across a fleet of commercial trucks, enabling the operations team to reduce idle-time incidents — directly supporting route optimisation decisions for field logistics. Evaluated clustering approaches across 15 real-world GPS datasets with 30K+ stoppage points using ARI, cluster-count drift, radius drift, and execution time; tuned BIRCH parameters to balance clustering quality, spatial stability, and scalability.
- **Stoppage Mining Module:** Developed a BigQuery-based stoppage mining pipeline applying 5 telemetry validity checks, Haversine-based distance correction, and POI-radius mapping to generate validated stoppage events; automated MySQL/CSV export with run-tracking and failure-alert logging.
- **MLOps / CI-CD Pipeline — Secure GCP Deployment:** Architected an automated GitHub Actions + GCP CI/CD pipeline with code quality checks, unit testing, Gitleaks, Secret Manager, and IAP-based VM deployment. Automated secure configuration delivery, Python environment setup, and 15-minute production data-processing jobs, replacing manual deployment and configuration workflows.
- **Fuel Analytics Algorithm Optimisation:** Partnered with the Fuel Analytics team to audit and retune telematics algorithms for fuel consumption calculation; improvements reduced signal-noise ratio in sensor data, increasing model accuracy and surfacing actionable fuel-efficiency insights for fleet managers.
- **Fuel Sensor Classification (Reed vs. Float) — 96% Accuracy:** Built and deployed an end-to-end GCP pipeline to classify vehicle fuel sensors using BigQuery telematics data; applied breakpoint detection and angle-based horizontal-segment analysis on fuel time-series. Validated predictions against a human-verified golden dataset and aggregated independent cycle-level predictions at vehicle and FERT-code levels using confidence-based voting, replacing manual inspection and running autonomously in production.

Advart Technologies | Data Analyst Intern **Dec 2024 – Feb 2025**
E-commerce Analytics / Python / Predictive Modelling / Dashboard Engineering

- **Sell-Out Forecasting Model:** Built a Python/Scikit-learn forecasting pipeline to predict daily product sell-out using lag, rolling-window, and calendar features; trained a Random Forest regression model with chronological validation and evaluated forecasts using MAE/RMSE, enabling monthly demand and GMV estimation for data-driven inventory planning.

Rabwin Industries | Data Analyst Intern **Jul 2024 – Dec 2024**
Power BI / Automated Reporting / EDA / Multi-source Data Integration

- **Reporting Efficiency +30%:** Automated a previously manual Power BI reporting workflow by integrating multi-system sales data into a single refreshable pipeline — cut report generation time by 30% and freed the sales team from weekly manual data pulls.

Technical Skills

Languages	Python (Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn)
Data & SQL	SQL, BigQuery, Complex Joins, CTEs, Window Functions, Aggregations, Geospatial SQL
ML / Data Science	Machine Learning, Deep Learning, Predictive Modelling, Time-Series Forecasting, Geospatial Analysis (HAC, BIRCH, Haversine, DBSCAN), EDA, Feature Engineering
MLOps & Cloud	GCP (Cloud Build, Cloud Scheduler, Secret Manager, Compute Engine, IAM, IAP, BigQuery), GitHub Actions, Gitleaks, CI/CD Pipelines, OIDC Keyless Auth, Docker (basics)
BI & Visualisation	Power BI, Interactive Python Dashboards (Plotly / Dash), Tableau (familiar)
Domain	Telematics, Fleet Analytics, Commercial Vehicle Data, IoT Sensor Data, E-commerce Analytics
Soft Skills	Cross-functional collaboration, Stakeholder communication, End-to-end problem ownership

Education

M.Sc. Data Science — Amrita Vishwa Vidyapeetham (Amrita University) 2020 – 2025
GPA: 7.5 / 10 | Coursework: Statistical Learning, Deep Learning, Big Data Systems, Machine Learning, Image Processing

Recognition

- **Appreciation Letter — VECV:** Received an appreciation letter from the Executive Vice President (EVP) of VECV for contributions to the Trip Detection Algorithm.